5. If X is the random variable with probability density function $f(x)$ given by,

$$f(x)=\begin{cases} x+1, & -1\le x<0 \\ -x+1, & 0\le x<1 \\ 0 & \text{otherwise} \end{cases}$$

then find (i) the distribution function $F(x)$ (ii) $P(-0.5\le X\le 0.5)$

6. If X is the random variable with distribution function $F(x)$ given by,

$$F(x)=\begin{cases} 0, & -\infty < x< 0 \\ \frac{1}{2}\left(x^2+x\right) & 0\le x<1 \\ 1, & 1\le x< \infty \end{cases}$$

then find (i) the probability density function $f(x)$ (ii) $P(0.3\le X\le 0.6)$

11.5 Mathematical Expectation

One of the important characteristics of a random variable is its expectation. Synonyms for expectation are expected value, mean, and first moment.

The definition of mathematical expectation is driven by conventional idea of numerical average. The numerical average of n numbers, say $a_1, a_2, a_3, \ldots, a_n$ is

$$\frac{a_1+a_2+a_3+\ldots+a_n}{n}\ .$$

The average is used to summarize or characterize the entire collection of n numbers $a_1, a_2, a_3, \ldots, a_n$, with single value.

Illustration 11.7

Consider ten numbers $6, 2, 5, 5, 2, 6, 2, -4, 1, 5$.

The average is $\dfrac{6+2+5+5+2+6+2-4+1+5}{10}=3$.

If ten numbers $6, 2, 5, 5, 2, 6, 2, -4, 1, 5$ are considered as the values of a random variable X the probability mass function is given by

x	-4	1	2	5	6
$P(X=x)$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{2}{10}$

The above calculation for average can also be rewritten as

$$-4\times\frac{1}{10}+1\times\frac{1}{10}+2\times\frac{3}{10}+5\times\frac{3}{10}+6\times\frac{2}{10}=3\ .$$

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